

01 Ordered Pairs & Cartesian Product

- $(a, b) = (c, d) \iff a = c \text{ and } b = d.$
- $A \times B = \{(a, b) : a \in A, b \in B\}.$
- $n(A \times B) = n(A) \cdot n(B).$
- $A \times B \neq B \times A$ in general (equal only if $A = B$ or one is empty).
- $A \times (B \cup C) = (A \times B) \cup (A \times C).$

02 Relations

A **relation** R from A to B is any subset of $A \times B$.

- Domain = $\{a : (a, b) \in R\}$; Range = $\{b : (a, b) \in R\}$; Codomain = B .
- Inverse: $R^{-1} = \{(b, a) : (a, b) \in R\}.$
- No. of relations from $A_{(m)}$ to $B_{(n)} = 2^{mn}.$
- No. of relations on a set of m elements = $2^{m^2}.$

03 Types of Relations

On a set A (with m elements):

Type	Condition
Reflexive	$(a, a) \in R \quad \forall a$
Symmetric	$(a, b) \in R \Rightarrow (b, a) \in R$
Transitive	$(a, b), (b, c) \in R \Rightarrow (a, c) \in R$
Antisymmetric	$(a, b), (b, a) \in R \Rightarrow a = b$
Equivalence	reflexive + symmetric + transitive

Reflexive relations: 2^{m^2-m} ; symmetric relations: $2^{m(m+1)/2}.$

04 Equivalence Relations & Classes

- An equivalence relation **partitions** A into disjoint **equivalence classes** $[a] = \{x \in A : xRa\}.$
- Any two classes are either equal or disjoint; their union is A .
- No. of equivalence relations on $A_{(m)} =$ Bell number $B_m.$

$B_1 = 1, B_2 = 2, B_3 = 5, B_4 = 15, B_5 = 52.$

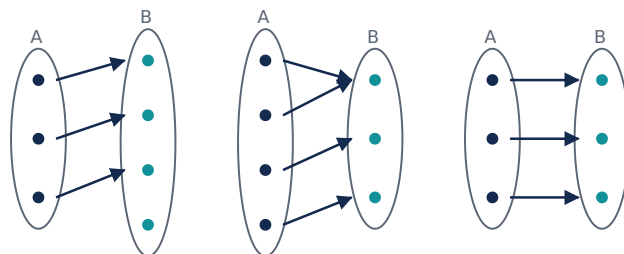
05 Functions: Definition

$f : A \rightarrow B$ assigns to **each** $a \in A$ **exactly one** $f(a) \in B$.

- Domain = A ; Codomain = B ; Range = $\{f(a) : a \in A\} \subseteq B.$
- **Vertical line test:** a graph is a function iff every vertical line meets it at most once.
- No. of functions $A_{(m)} \rightarrow B_{(n)} = n^m.$

06 Types of Functions

$f : A \rightarrow B$ classified as:



one-one, into

onto (many-one)

bijective

- **One-one** (injective): $f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$
- **Onto** (surjective): range = codomain; else **into**.
- **Bijective:** one-one and onto.
- **Horizontal line test:** one-one iff every horizontal line meets the graph at most once.

07 Counting Functions

From $A_{(m)}$ to $B_{(n)}$:

Kind	Count
All functions	n^m
Injective ($m \leq n$)	${}_n P_m = \frac{n!}{(n-m)!}$ (else 0)
Surjective ($m \geq n$)	$\sum_{r=0}^n (-1)^r {}^n C_r (n-r)^m$
Bijective ($m = n$)	$n!$

08 Standard Functions

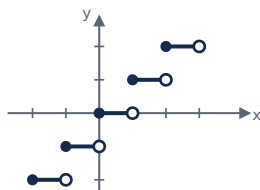
Function	Rule
Identity	$I(x) = x$
Constant	$f(x) = c$
Polynomial	$a_n x^n + \dots + a_1 x + a_0$
Rational	$\frac{p(x)}{q(x)}, q(x) \neq 0$
Modulus / Signum	$ x $; $\text{sgn}(x)$
Exponential / Log	a^x ; $\log_a x$ ($a > 0, a \neq 1$)

09 Domain & Range (Standard)

$f(x)$	Domain	Range
x^2	$\mathbb{R}$	$[0, \infty)$
$\sqrt{x}$	$[0, \infty)$	$[0, \infty)$
$\frac{1}{x}$	$\mathbb{R} \setminus \{0\}$	$\mathbb{R} \setminus \{0\}$
$ x $	$\mathbb{R}$	$[0, \infty)$
$[x]$	$\mathbb{R}$	$\mathbb{Z}$
$\{x\}$	$\mathbb{R}$	$[0, 1)$
e^x	$\mathbb{R}$	$(0, \infty)$
$\log x$	$(0, \infty)$	$\mathbb{R}$
$\sin x, \cos x$	$\mathbb{R}$	$[-1, 1]$

10 Greatest Integer & Fractional Part

$[x]$ = greatest integer $\leq x$; $\{x\} = x - [x]$.



$y = [x]$: constant on $[n, n+1)$, jumps +1 at each integer (● included, ○ excluded).

- $[x] \leq x < [x] + 1$; $0 \leq \{x\} < 1$.
- $[x + n] = [x] + n$ ($n \in \mathbb{Z}$);

$$[x] + [-x] = \begin{cases} 0 & x \in \mathbb{Z} \\ -1 & \text{else} \end{cases}$$
- Hermite:

$$[x] + [x + \frac{1}{n}] + \dots + [x + \frac{n-1}{n}] = [nx].$$

11 Modulus & Signum

- $|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$
 $|x| \geq 0$, $\sqrt{x^2} = |x|$, $|x|^2 = x^2$.
- $|xy| = |x||y|$; $|x+y| \leq |x| + |y|$;
 $||x| - |y|| \leq |x - y|$.
- $|x| < a \iff -a < x < a$;
 $|x| > a \iff x < -a \text{ or } x > a$ ($a > 0$).
- $\text{sgn}(x) = \frac{x}{|x|}$ ($x \neq 0$), $\text{sgn}(0) = 0$.

12 Algebra of Functions

For real functions f, g :

- $(f \pm g)(x) = f(x) \pm g(x)$ and
 $(fg)(x) = f(x)g(x)$ — domain
 $= \text{dom } f \cap \text{dom } g$.
- $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ — domain
 $= \text{dom } f \cap \text{dom } g \cap \{x : g(x) \neq 0\}$.

13 Composite Functions

- $(g \circ f)(x) = g(f(x))$; domain
 $= \{x \in \text{dom } f : f(x) \in \text{dom } g\}$.
- Associative:** $f \circ (g \circ h) = (f \circ g) \circ h$.
- Not commutative:** $f \circ g \neq g \circ f$ in general.
- f, g injective $\Rightarrow g \circ f$ injective; both surjective $\Rightarrow$ surjective.
- $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

14 Inverse Functions

- f^{-1} exists $\iff f$ is **bijective**.
- $(f^{-1} \circ f)(x) = x$ and $(f \circ f^{-1})(y) = y$.
- Graph of f^{-1} is the reflection of f in the line $y = x$.
- $(f^{-1})^{-1} = f$; f increasing $\Rightarrow f^{-1}$ increasing.
- To find: solve $y = f(x)$ for x , then interchange x, y .

15 Even & Odd Functions

- Even:** $f(-x) = f(x)$ — symmetric about the y -axis ($x^2, \cos x, |x|$).
- Odd:** $f(-x) = -f(x)$ — symmetric about the origin, $f(0) = 0$ ($x^3, \sin x, \tan x$).
- Any f splits uniquely:

$$f(x) = \underbrace{\frac{f(x)+f(-x)}{2}}_{\text{even}} + \underbrace{\frac{f(x)-f(-x)}{2}}_{\text{odd}}$$
- even $\times$ even = even, odd $\times$ odd = even, even $\times$ odd = odd.

16 Periodic Functions

- $f(x+T) = f(x) \forall x$; the least positive T is the **fundamental period**.
- $\sin x, \cos x : 2\pi$; $\tan x, \cot x : \pi$;
 $|\sin x|, |\cos x| : \pi$; $\{x\} : 1$.
- $f(ax+b)$ has period $\frac{T}{|a|}$.
- Period of $f \pm g = \text{LCM}(T_1, T_2)$ when it exists.

17 Graphical Transformations

Change	Effect on graph
$f(x) + c$ / $f(x) - c$	shift up / down by c
$f(x + c)$ / $f(x - c)$	shift left / right by c
$a f(x)$	vertical stretch $ a > 1$ / compress
$f(ax)$	horizontal compress $ a > 1$ / stretch
$-f(x)$ / $f(-x)$	reflect in x -axis / y -axis
$ f(x) $	flip the part below the x -axis upward
$f(x)$	mirror the $x \geq 0$ part to $x < 0$

18 Functional Equations

Relation (for all x, y)	Solution
$f(x + y) = f(x) + f(y)$	$f(x) = kx$
$f(x + y) = f(x) f(y)$	$f(x) = a^x$
$f(xy) = f(x) + f(y)$	$f(x) = k \log_a x$
$f(xy) = f(x) f(y)$	$f(x) = x^n$
$f(x) f(\frac{1}{x}) = f(x) + f(\frac{1}{x})$	$f(x) = 1 \pm x^n$

(Continuity or a mild regularity assumption is understood.)

19 Quick Recap

- $n(A \times B) = n(A) n(B)$; relations $A \rightarrow B : 2^{mn}$.
- Functions $A_{(m)} \rightarrow B_{(n)} : n^m$; injective ${}^n P_m$; bijective $n!$.
- f^{-1} exists $\iff$ bijective; graph reflects in $y = x$.
- Even $f(-x) = f(x)$; odd $f(-x) = -f(x)$.
- $g \circ f \neq f \circ g$ in general — order matters.

20 Common Traps

- ! A relation is a function only if **every** domain element has **exactly one** image.
- ! Onto depends on the **codomain** — shrinking it can make a function onto.
- ! $[x]$ is **not** rounding: $[-2.3] = -3$, not -2 .
- ! $f \circ g \neq g \circ f$ — composition order matters.
- ! Inverse needs a **bijection**: restrict domain/codomain first (e.g. **sin** on $[-\frac{\pi}{2}, \frac{\pi}{2}]$).